Expert labels across the two gate states

Implementation schematic | R4: DART-style + gate / R6: DAgger + gate / R8: DART-style + DAgger + gate

$g(o_t) = 1$ **iff** $|d_t| < 0.12$ **m and** $|h_t| < 0.075$ **rad; otherwise** $g(o_t) = 0$**.**

$d_t =$ planar target distance − height-dependent standoff; $h_t =$ target heading error.

Gate is recomputed at every step: 0 ↔ 1.

One 22-input MLP; gate is an input, not a hard output switch.

Executed action

Teacher supervision (saved label)

All collectors label each pre-action state at 1000 Hz.

A DART-style

R4: teacher-driven noisy collection

B DAgger

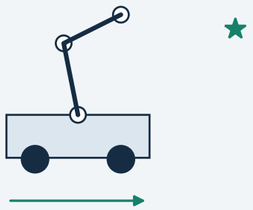
R6: student-driven collection

C DART-style + DAgger

R8: DART-initialised student collection

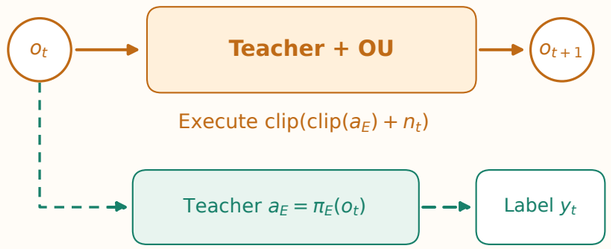
01 GATE = 0

Navigate + arm posture



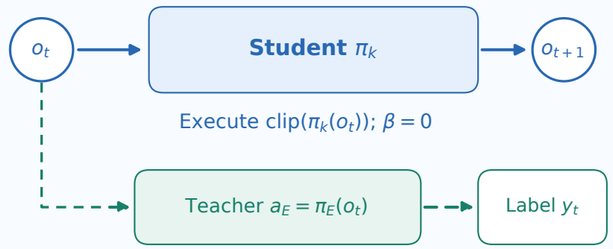
Teacher arm: fixed-pose PD

State reached by teacher + OU noise



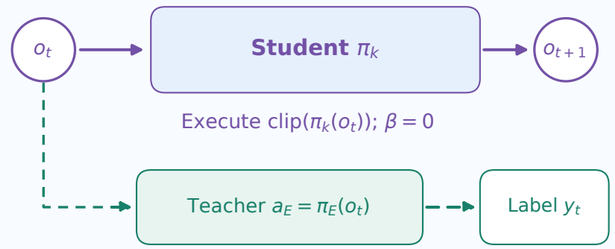
Noise: arm + wheels before first gate activation.
Clean labels teach recovery from noisy states.

State reached by the current student



Label: wheel alignment / standoff correction;
arm torque restores the fixed hold posture.

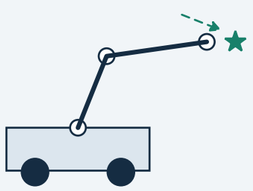
State reached by the current student



Label: wheel alignment / standoff correction;
arm torque restores the fixed hold posture.

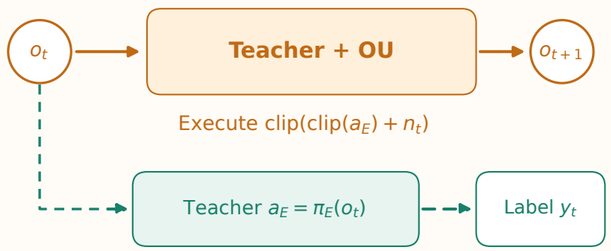
02 GATE = 1

Arm reach + hold



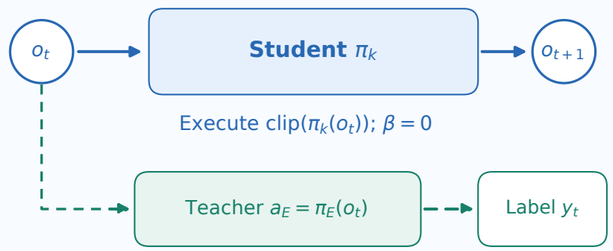
Teacher arm: Cartesian reach

State reached by teacher + OU noise



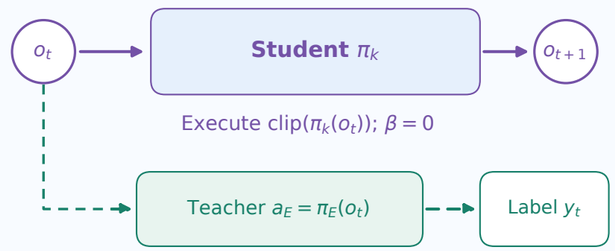
Wheel noise has latched OFF; arm OU continues.*
Clean labels teach recovery during reaching.

State reached by the current student



Label: arm torque corrects reaching / hold error;
wheel labels still correct residual base errors.

State reached by the current student



Label: arm torque corrects reaching / hold error;
wheel labels still correct residual base errors.

WHEN THE LABEL TAKES EFFECT

Saved target → loss
→ updated policy

No teacher takeover
during DAgger rollout.

OFFLINE TRAINING

$D_{\text{Clean}} \cup D_{\text{DART}}$

→ Train student on clean teacher labels

DART data: sustained-hold passes only.
Perturbations are configured OU noise.

OFFLINE TRAINING

$D_{\text{Clean}} \cup \bigcup_{i=1}^{k} D_{\text{DAgger},i}$

→ Train π_{k+1} from scratch; collect again

Failed student episodes are retained.
Teacher labels change the NEXT round's policy.

OFFLINE TRAINING

$D_{\text{Clean}} \cup D_{\text{DART}} \cup \bigcup_{i=1}^{k} D_{\text{DAgger},i}$

→ Train π_{k+1} from scratch; collect again

DAgger rollouts add NO OU noise.
Both data sources supervise the next policy.

* DART noise: wheels latch off at the first g=1 (even if g later returns to 0); all noise stops 1.5 s after first task reach.

Gate=1 can include base micro-adjustments. DAgger excludes numerical-failure / empty episodes. Schematic: no measured trajectories or performance claims.